

Omar Gamal

Software Engineer & Game Engine Developer

📍 Heliopolis, Cairo, Egypt

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🔗 LinkedIn Profile

📄 GitHub Profile

Professional Summary

Passionate Software Engineer with 7+ years of experience in full-stack development, game engine architecture, and plugin development. Proven track record of building scalable applications, custom game engines, and AI-powered solutions. Started programming at age 10 and evolved into a senior developer specializing in systems programming, 3D graphics, and enterprise web applications. Strong background in DirectX11, ECS architecture, and modern web technologies.

Technical Skills

Programming Languages

C++ Java JavaScript
Python C Go

Web Technologies

React.js Next.js
Node.js Express.js
Gin Fiber HTML5
CSS3 REST APIs
Nginx TypeScript

DevOps & Infrastructure

Docker Kubernetes
Linux WSL
GitHub Actions CI/CD
Git

Testing & Quality Assurance

Jest Vitest
React Testing Library
Cypress Testify
GoMock Unit Testing
Integration Testing

Game Development

DirectX11 Unity
Unreal Engine
ECS Architecture
3D Graphics
Game Engines

Databases & Caching

MongoDB MySQL
PostgreSQL Redis
Database Administration

Professional Experience

Senior Spigot Plugin Developer

ATTILA GAMING

January 2022 - Present | Part-time

Lead development of custom Minecraft plugins using Java and Spigot API. Architected scalable solutions for high-traffic gaming servers, implementing performance optimizations that improved server response times by 40%. Designed complex gameplay mechanics and automated systems serving 1000+ concurrent players.

Lead Developer

Minecraft Gaming Servers

2018 - 2022

Promoted to Lead Developer role on major Minecraft server networks. Managed development team and technical architecture decisions. Implemented monetization features and server optimization strategies that increased revenue and player retention. Gained extensive experience in Java enterprise patterns and distributed systems.

Freelance Full-Stack Developer

Self-Employed

May 2018 - January 2022

Delivered 20+ web applications and AI chatbots for diverse clients across multiple industries. Specialized in React.js, Node.js, and MongoDB stack. Developed responsive websites with optimal performance and SEO optimization. Created intelligent chatbots using natural language processing and machine learning algorithms.

Key Projects

NexusCommerce - Enterprise E-Commerce Platform *(In Development)*

React.js, TypeScript, Express.js, Go, Gin, PostgreSQL, Redis, Docker, Kubernetes, Jest, Testify

Developing a scalable microservices-based e-commerce platform showcasing full-stack expertise. Features dual backend architecture with Express.js for rapid development and Go with Gin framework for high-performance services. Implements comprehensive CI/CD pipeline with Jest and Testify for automated testing suites, and Kubernetes orchestration for production-ready scalability. Includes user authentication, payment processing, inventory management, and real-time analytics.

 **Repository:** [View Source Code](#)

NebulaEngine - Next-Generation Game Engine *(In Development)*

C++, DirectX11/12, Vulkan, Cross-Platform Architecture

Currently developing a cutting-edge 2D/3D game engine with multi-platform support and modern graphics API integration. Features include advanced ECS system, memory optimization, and planned visual editor. Architecting NebulaSync library for C++ reflection and network synchronization. Focus on performance optimization and developer-friendly API design.

 **Repository:** [View Source Code](#)

DarkFire Game Engine

C++, DirectX11, ECS Architecture, Windows API

Built complete 3D game engine from scratch using DirectX11 API. Implemented full Entity Component System (ECS) architecture with memory optimization. Developed space shooter game as proof of concept, demonstrating engine capabilities including 3D rendering, physics, and game loop management.

 **Repository:** [View Source Code](#)

AI & Machine Learning Models

Python, Neural Networks, Game AI

Researched and implemented various machine learning models for game development applications. Studied neural network architectures and different learning paradigms. Applied AI concepts to create intelligent game behaviors and decision-making systems.

Portfolio Website

Next.js, React.js, Modern Web Technologies

Developed responsive portfolio website showcasing projects and technical expertise. Implemented modern web development practices with optimal performance and SEO. Features dynamic content management and responsive design principles.

 **Live Site:** [Visit Portfolio](#)

 **Source Code:** [View Source Code](#)

Key Achievements

- ▶ Self-taught programming from age 10, demonstrating exceptional learning ability and passion for technology
- ▶ Successfully monetized Minecraft plugin development, earning revenue as lead developer
- ▶ Architected and built two complete game engines from scratch using advanced C++ and graphics programming
- ▶ Delivered 20+ successful freelance projects with 100% client satisfaction rate
- ▶ Gained expertise in both low-level systems programming and high-level web development
- ▶ Contributed to server optimization projects serving 1000+ concurrent users
- ▶ Developed cross-platform solutions supporting multiple operating systems and graphics APIs

Education

- Bachelor of Science in Computer Science**
Helwan University, Cairo
Expected Graduation: March 2026 | Currently in Third Year

GPA: 3.93/4.0 - Specialization in Software Engineering with focus on advanced algorithms, data structures, and software architecture patterns. Consistently maintained Dean's List status while balancing full-time studies with professional development work.